

Software Requirements Specification & Product Module Specification

Robotics & Electronics Commerce Platform

Production-grade e-commerce platform for robotics, electronics, maker components, development boards, tools and related services.

Document	SRS + Product / Module Specification
Version	1.0
Status	Product development baseline
Audience	Founders, Product, Engineering, QA, Design, DevOps, Operations
Primary market	India; architecture prepared for future expansion

This specification defines product behavior and implementation requirements. It is inspired by common electronics-commerce functionality and does not reproduce proprietary branding, copy, images or implementation from an existing retailer.

1. Executive Summary and Product Vision

The product is a modern online store for robotics and electronics. Customers can discover technically detailed products, compare alternatives, purchase individual or bulk quantities, pay securely, track delivery, request returns/refunds, and obtain technical support.

The platform is designed as a commercial system: reliable checkout, accurate inventory, auditable administration, SEO-friendly product pages, structured technical specifications, GST-aware invoicing, and an architecture that can evolve toward B2B and multi-warehouse operations.

Product principles

- Technical accuracy: product specifications, compatibility and documentation are first-class data.
- Trustworthy commerce: price, stock, tax, payment and order state are server-authoritative.
- Fast discovery: search, filtering, category navigation and comparison reduce time-to-product.
- Operational control: staff use granular permissions, workflows and audit trails.
- Scalable foundation: start as a modular monolith and extract services only when justified.
- India-ready: INR, GST, UPI/payment workflows and shipping integrations are core considerations.

2. Goals and Success Metrics

Goal	Indicative success metric
Product discovery	Strong search/category-to-product engagement
Conversion	Improving product-view to checkout and checkout to paid-order conversion
Checkout reliability	Very low payment/order creation failure rate
Inventory accuracy	Minimal overselling and reconciliation exceptions
Customer satisfaction	Low avoidable return/support rate; strong review sentiment
Operations	Reduced manual catalog/order work
Performance	Strong Core Web Vitals and predictable API latency

Exact targets should be finalized from business baseline data after launch; the table defines measurement categories rather than contractual numbers.

3. Scope and Assumptions

In scope: B2C storefront, product/catalog management, commerce transactions, payments, shipping, customer accounts, reviews, support, administration, analytics, GST/invoicing and B2B foundations.

Initial exclusions: full third-party marketplace seller onboarding, international tax engines, native mobile apps and advanced ML recommendations.

Assumptions: the business owns or licenses product data/media; providers expose supported APIs/webhooks; legal/tax policies are reviewed by qualified professionals; infrastructure is cloud-hosted.

4. Personas, Actors and Roles

Role	Primary responsibilities
Guest	Browse, search, compare and add to cart
Customer	Purchase, account, orders, reviews, returns and support
B2B Customer	Bulk pricing, GST/business profile, RFQ and purchase-order workflows
Catalog Manager	Products, categories, brands, attributes, media and content
Inventory Manager	Stock, reservations, adjustments and warehouse operations
Order Manager	Order processing, shipment, cancellation, returns and refunds
Support Agent	Customer tickets, order issues and technical questions
Finance/Admin	Payments, invoices, tax configuration, refunds and reports
Marketing Manager	Campaigns, coupons, banners and merchandising
Super Admin	System configuration, roles, permissions and security-sensitive actions

5. Recommended Technology Stack and Architecture

Layer	Recommended technology
Web	Next.js + TypeScript
UI	Tailwind CSS + shadcn/ui
Backend	NestJS + TypeScript
Database	PostgreSQL + Prisma
Cache/queues	Redis + BullMQ
Search	PostgreSQL initially; OpenSearch when justified
Object storage	Cloudflare R2 or AWS S3
Payments	Razorpay or another compliant payment provider
Shipping	Provider abstraction
Runtime	Docker
CI/CD	GitHub Actions
Edge/CDN/WAF	Cloudflare
Monitoring	Sentry + OpenTelemetry
Analytics	PostHog and/or GA4
Repository	Turborepo + pnpm

Architecture recommendation: begin as a modular monolith with explicit domain boundaries. High-load domains such as search, notifications or fulfillment can later be extracted without redesigning the whole platform.

Logical flow: Browser -> Next.js -> NestJS API -> PostgreSQL / Redis / Object Storage. Background workers process asynchronous jobs. Payment, shipping, notification and analytics providers integrate through adapters.

6. Functional Requirements

FR-PLAT-001 - Maintain server-side source of truth for product, price, inventory, order and payment state.

FR-PLAT-002 - Assign unique identifiers to users, products, SKUs, orders, payments, shipments, returns and tickets.

FR-PLAT-003 - Record material administrative changes in an append-only or otherwise tamper-evident audit trail.

FR-PLAT-004 - Enforce role-based access control for administrative and operational functions.

FR-STORE-001 - Expose SEO-friendly category, brand and product pages.

FR-CAT-001 - Support categories, brands, products, SKUs, variants and technical attributes.

FR-SEARCH-001 - Search by product names, SKU, brand, category and relevant technical attributes.

FR-INV-001 - Distinguish on-hand, reserved and available inventory.

FR-CART-001 - Add, remove and update cart items subject to stock and limits.

FR-CHK-001 - Validate address, stock, price, tax, shipping and payment before order confirmation.

FR-PAY-001 - Verify payment state server-side and reconcile provider webhooks.

FR-ORD-001 - Maintain a controlled order lifecycle state machine.

FR-SHIP-001 - Create and track shipments through an integration abstraction.

FR-RET-001 - Support return, replacement and refund workflows according to configured policy.

FR-REV-001 - Support ratings, reviews, moderation and verified-purchase indicators.

FR-NOT-001 - Send transactional notifications asynchronously.

FR-B2B-001 - Support business profiles, GST details, bulk pricing and RFQ foundations.

FR-TAX-001 - Store tax components needed for applicable Indian invoicing workflows.

FR-ANL-001 - Capture stable business events for reporting and product analytics.

MOD-01 - Storefront / Customer Experience

Objective: Provide the public shopping experience from discovery through purchase and post-purchase support.

Actors: Guest, Customer, B2B Customer

Core features:

- Responsive home page
- Navigation/mega-menu
- Category/brand pages
- Product detail
- Comparison
- Account
- Order tracking

Functional requirements:

FR-STORE-101 - Home page supports configurable promotional sections without code deployment.

FR-STORE-102 - Product pages show price, stock, SKU, specifications, media, documents, reviews and related products.

FR-STORE-103 - Storefront supports mobile, tablet and desktop layouts.

FR-STORE-104 - Critical purchase actions remain usable under slow network conditions.

Business rules: Merchandising content is role-controlled; price and stock are sourced from authoritative APIs.

Dependencies: Catalog, pricing, inventory, CMS, search

Edge cases: Missing media; discontinued product; price change; item unavailable during checkout.

Acceptance criteria: Users can navigate, search, inspect technical details and add an available SKU to cart.

MOD-02 - Authentication & User Management

Objective: Manage identity, account data, sessions and customer/business profiles.

Actors: Guest, Customer, Admin, Staff

Core features:

- Registration/login
- Password reset
- Verification
- Profile
- Addresses
- Sessions
- Roles

Functional requirements:

FR-AUTH-001 - Support secure registration and login.

FR-AUTH-002 - Never store passwords in plaintext.

FR-AUTH-003 - Protect sensitive account actions with re-authentication or verification.

FR-AUTH-004 - Allow authorized admins to suspend/disable accounts.

Business rules: Account identifiers are unique; protected APIs enforce authorization server-side.

Dependencies: Identity store, email/SMS, RBAC

Edge cases: Duplicate identity; expired OTP; revoked session; disabled account.

Acceptance criteria: Valid users authenticate and cannot access resources outside their authorization.

MOD-03 - Product Catalog, SKU & Specifications

Objective: Provide the canonical technical and commercial representation of products.

Actors: Catalog Manager, Admin, Customer

Core features:

- Products
- SKUs
- Variants
- Attributes
- Technical specs
- Media
- Datasheets
- Compatibility
- Related products

Functional requirements:

FR-CAT-101 - A product supports one or more SKUs.

FR-CAT-102 - SKU is the inventory and purchasable unit identifier.

FR-CAT-103 - Specifications use flexible category-aware attributes.

FR-CAT-104 - Catalog edits are auditable.

FR-CAT-105 - Documents have metadata and access controls.

Business rules: SKU is unique; archived products remain referenceable by historical orders.

Dependencies: Categories, brands, inventory, object storage, pricing

Edge cases: Duplicate SKU; conflicting attribute types; deleted variant with order history.

Acceptance criteria: Managers can create, validate, publish and search a complete product.

MOD-04 - Categories & Brands

Objective: Organize products for navigation, merchandising and search.

Actors: Catalog Manager, Admin

Core features:

- Hierarchical categories
- Brand pages
- SEO metadata
- Ordering
- Visibility

Functional requirements:

FR-CAT-201 - Categories support parent-child relationships.

FR-CAT-202 - Categories and brands support SEO metadata.

FR-CAT-203 - Authorized users can reorder visible categories.

Business rules: Prevent circular category relationships; block or archive deletion when dependencies exist.

Dependencies: Catalog, CMS, search

Edge cases: Orphaned products; category cycle; hidden indexed category.

Acceptance criteria: Valid category trees and brand/category paths remain navigable.

MOD-05 - Search, Filter & Discovery

Objective: Enable fast and relevant product discovery.

Actors: Guest, Customer, B2B Customer

Core features:

- Search
- Autocomplete
- Filters
- Sort
- Facets
- Synonyms
- Typo tolerance

Functional requirements:

FR-SEARCH-101 - Return relevant results for product names and SKU queries.

FR-SEARCH-102 - Derive filters from applicable attributes.

FR-SEARCH-103 - Update indexes after catalog changes.

FR-SEARCH-104 - Record search events.

Business rules: Unavailable products may be hidden or deprioritized by merchandising policy.

Dependencies: Catalog, search index, analytics

Edge cases: Typo/no-result query; stale index; conflicting filters.

Acceptance criteria: Common searches return useful results and filters narrow results correctly.

MOD-06 - Inventory & Warehouse

Objective: Maintain accurate stock and prevent overselling.

Actors: Inventory Manager, Order Manager, Admin

Core features:

- On-hand
- Reserved
- Available
- Adjustments
- Low-stock alerts
- Warehouse readiness
- Transfers

Functional requirements:

FR-INV-101 - Available stock equals on-hand minus active reservations and configured deductions.

FR-INV-102 - Checkout reserves stock using a concurrency-safe mechanism.

FR-INV-103 - Stock adjustments require a reason and audit record.

FR-INV-104 - Inventory model supports warehouse/location identifiers.

Business rules: Negative stock is prohibited except for explicitly controlled exceptional workflows.

Dependencies: Orders, products, database transactions, Redis

Edge cases: Concurrent checkout; expired reservation; cancellation; partial fulfillment.

Acceptance criteria: Concurrent purchases cannot oversell under configured rules and all adjustments are traceable.

MOD-07 - Cart & Wishlist

Objective: Persist shopping intent and saved products.

Actors: Guest, Customer

Core features:

- Cart items
- Quantity controls
- Wishlist
- Move to cart
- Price/stock revalidation

Functional requirements:

FR-CART-101 - Users can add/remove valid SKUs.

FR-CART-102 - Cart totals are recalculated server-side.

FR-CART-103 - Price and stock are revalidated before checkout.

FR-CART-104 - Wishlist persists for authenticated users.

Business rules: Cart contents do not guarantee inventory until reservation succeeds.

Dependencies: Catalog, pricing, inventory, authentication

Edge cases: Price change; discontinued SKU; quantity limit; guest-to-user merge.

Acceptance criteria: Cart reflects current authoritative pricing and availability.

MOD-08 - Checkout

Objective: Convert a validated cart into an order/payment attempt.

Actors: Customer, B2B Customer

Core features:

- Address
- Shipping
- Tax
- Coupon
- Summary
- Payment initiation

Functional requirements:

FR-CHK-101 - Validate every line item against current catalog and inventory.

FR-CHK-102 - Calculate final payable amount server-side.

FR-CHK-103 - Prevent duplicate order creation for repeated client requests.

FR-CHK-104 - Failed payments cannot create falsely paid orders.

Business rules: Use idempotency for order/payment initiation.

Dependencies: Cart, inventory, pricing, tax, coupon, payment, shipping

Edge cases: Double click; refresh; timeout; coupon invalidated.

Acceptance criteria: Valid checkout creates one coherent order/payment attempt.

MOD-09 - Payments

Objective: Integrate secure payment processing and reconciliation.

Actors: Customer, Finance/Admin

Core features:

- Payment intent
- Provider integration
- Webhook handling
- Verification
- Refunds
- Reconciliation

Functional requirements:

FR-PAY-101 - Payment attempts have unique identifiers.

FR-PAY-102 - Verify provider webhook signatures/events where supported.

FR-PAY-103 - Reconcile payment status before marking an order paid.

FR-PAY-104 - Record refund amount, reason and provider reference.

Business rules: Never treat a browser-only success callback as proof of payment.

Dependencies: Payment provider, orders, finance

Edge cases: Late/duplicate webhook; partial refund; provider outage.

Acceptance criteria: Successful payments reconcile correctly and duplicate events are idempotent.

MOD-10 - Orders

Objective: Manage the complete commercial order lifecycle.

Actors: Customer, Order Manager, Finance, Support

Core features:

- Order creation
- State machine
- Order items
- Invoices
- Cancellation
- History

Functional requirements:

FR-ORD-101 - Every order has a unique human-readable order number.

FR-ORD-102 - Transitions are validated against allowed states.

FR-ORD-103 - Historical price/tax/item data is preserved.

FR-ORD-104 - Customers can view order history.

Business rules: Historical orders are immutable except through controlled correction workflows.

Dependencies: Checkout, payment, inventory, shipping, invoice

Edge cases: Partial cancellation; split shipment; payment reversal.

Acceptance criteria: Orders follow valid transitions and preserve commercial history.

MOD-11 - Shipping & Delivery

Objective: Create, track and reconcile shipments.

Actors: Order Manager, Customer

Core features:

- Serviceability
- Rates
- Shipment creation
- Tracking
- Delivery events
- Returns

Functional requirements:

FR-SHIP-101 - Abstract courier-provider operations.

FR-SHIP-102 - Store provider and tracking references.

FR-SHIP-103 - Process tracking events idempotently.

FR-SHIP-104 - Customers see current shipment status.

Business rules: Provider-specific states map to a platform-wide vocabulary.

Dependencies: Orders, shipping provider, notifications

Edge cases: Provider outage; duplicate event; RTO; split shipment.

Acceptance criteria: Shipment creation and tracking work through a configured provider.

MOD-12 - Returns, Replacements & Refunds

Objective: Handle post-purchase exceptions according to policy.

Actors: Customer, Support, Order Manager, Finance

Core features:

- Return request
- Eligibility
- Approval
- Pickup
- Inspection
- Replacement
- Refund

Functional requirements:

FR-RET-101 - Evaluate return eligibility against configurable policy rules.

FR-RET-102 - Record reason and evidence when required.

FR-RET-103 - Refunds reference originating order/payment.

FR-RET-104 - Replacement orders link to the original order.

Business rules: Policy can vary by category and product condition.

Dependencies: Orders, shipping, payments, inventory, support

Edge cases: Partial return; damaged item; failed pickup.

Acceptance criteria: Eligible returns follow controlled logistics and financial workflows.

MOD-13 - Reviews, Ratings & Product Q&A;

Objective: Collect trustworthy customer feedback and technical questions.

Actors: Customer, Support, Catalog Manager, Admin

Core features:

- Ratings
- Reviews
- Images
- Verified purchase
- Moderation
- Q&A;

Functional requirements:

FR-REV-101 - Eligible customers can review purchased products.

FR-REV-102 - Moderation status is separate from review content.

FR-REV-103 - Verified purchase status derives from order evidence.

FR-REV-104 - Users can report inappropriate content.

Business rules: Staff edits/moderation are auditable.

Dependencies: Orders, users, catalog, moderation

Edge cases: Duplicate review; abusive content; deleted product.

Acceptance criteria: Eligible feedback is submitted, moderated and displayed correctly.

MOD-14 - Notifications

Objective: Deliver transactional and operational messages asynchronously.

Actors: Customer, Staff, System

Core features:

- Email
- SMS
- Push/WhatsApp readiness
- Templates
- Preferences
- Retries

Functional requirements:

FR-NOT-101 - Transactional messages originate from domain events.

FR-NOT-102 - Jobs support retry with backoff.

FR-NOT-103 - Delivery attempts are recorded.

FR-NOT-104 - Marketing respects consent/preferences.

Business rules: Critical transaction messages do not depend on the browser remaining open.

Dependencies: Orders, payments, shipping, providers, BullMQ

Edge cases: Provider outage; duplicate event; invalid recipient.

Acceptance criteria: Important messages are queued, retried safely and traceable.

MOD-15 - Customer Support

Objective: Provide structured support for orders, products and technical issues.

Actors: Customer, Support Agent, Admin

Core features:

- Tickets
- Categories
- Priority
- Assignment
- Internal notes
- Attachments
- SLA tracking

Functional requirements:

FR-SUP-101 - Customers can create support tickets.

FR-SUP-102 - Tickets support status, priority and assignment.

FR-SUP-103 - Internal notes are not customer-visible.

FR-SUP-104 - Tickets can link to orders/products/payments.

Business rules: Minimize and protect sensitive personal information.

Dependencies: Users, orders, products, notifications

Edge cases: Duplicate ticket; escalated ticket; attachment failure.

Acceptance criteria: Agents manage tickets while customers see only authorized content.

MOD-16 - Coupons, Pricing & Promotions

Objective: Manage commercial pricing rules and promotions.

Actors: Marketing Manager, Admin, Customer

Core features:

- Coupons
- Percentage/fixed discounts
- Minimum order
- Targeting
- Usage limits
- Schedules

Functional requirements:

FR-PROMO-101 - Coupon eligibility is evaluated server-side.

FR-PROMO-102 - Usage limits are concurrency-safe.

FR-PROMO-103 - Promotion rules are auditable and time-bounded.

FR-PROMO-104 - Checkout explains invalid/expired coupons safely.

Business rules: Promotion stacking must be explicitly configured.

Dependencies: Catalog, cart, checkout, orders

Edge cases: Race on last use; expired campaign; conflicting promotions.

Acceptance criteria: Only eligible orders receive discounts and usage counts remain correct.

MOD-17 - CMS & Content

Objective: Manage non-transactional content supporting SEO, education and merchandising.

Actors: Content Manager, Marketing, Admin

Core features:

- Banners
- Landing pages
- Blogs
- Tutorials
- Buying guides
- FAQs
- SEO metadata

Functional requirements:

FR-CMS-101 - Authorized users can publish/unpublish content.

FR-CMS-102 - Content supports draft/published states.

FR-CMS-103 - SEO title, description and canonical metadata are configurable.

FR-CMS-104 - Media references use managed object storage.

Business rules: Published content must pass validation for required metadata.

Dependencies: Object storage, storefront, analytics

Edge cases: Broken media; duplicate slug; unpublished linked page.

Acceptance criteria: Content can be created, previewed and published without code deployment.

MOD-18 - Marketing & Merchandising

Objective: Improve discovery and conversion through controlled merchandising.

Actors: Marketing Manager, Admin

Core features:

- Featured products
- Campaigns
- Banners
- Cross-sell
- Upsell
- Abandoned cart
- Referral readiness

Functional requirements:

FR-MKT-101 - Merchandising rules support date ranges and priority.

FR-MKT-102 - Marketing actions are measurable through analytics events.

FR-MKT-103 - Abandoned-cart campaigns exclude completed/cancelled orders.

Business rules: Marketing cannot override hard inventory or compliance constraints.

Dependencies: CMS, catalog, analytics, notifications

Edge cases: Out-of-stock featured item; overlapping campaigns.

Acceptance criteria: Configured campaigns render correctly and generate measurable events.

MOD-19 - Analytics & Reporting

Objective: Provide operational and product insight.

Actors: Admin, Finance, Marketing, Product

Core features:

- Revenue
- Orders
- AOV
- Conversion
- Product performance
- Inventory turnover
- Search analytics

Functional requirements:

FR-ANL-101 - Core commerce events have stable names and properties.

FR-ANL-102 - Reports use consistent definitions for revenue, orders and refunds.

FR-ANL-103 - Unnecessary sensitive data is excluded from analytics tools.

Business rules: Financial reports reconcile to authoritative transactional data.

Dependencies: Orders, payments, inventory, analytics

Edge cases: Late/duplicate events; timezone differences.

Acceptance criteria: Dashboards reconcile to source data and event definitions are documented.

MOD-20 - GST, Tax & Invoicing

Objective: Support India-oriented tax and invoice requirements with configurable rules.

Actors: Finance, Admin, Customer, B2B Customer

Core features:

- GSTIN
- HSN/SAC
- CGST/SGST/IGST
- Tax invoice
- Credit note
- Refund records

Functional requirements:

FR-TAX-101 - Store tax components applied to each order.

FR-TAX-102 - Invoices preserve historical seller/customer/tax data.

FR-TAX-103 - Business customers can store GST information subject to policy.

FR-TAX-104 - Tax logic is centralized and testable.

Business rules: Tax logic and legal compliance must be reviewed by qualified Indian tax/accounting professionals before production.

Dependencies: Orders, addresses, product tax classification, finance

Edge cases: Interstate/intrastate; invalid GSTIN; special treatment.

Acceptance criteria: Invoice totals reconcile to order totals and tax components are traceable.

MOD-21 - B2B, Bulk Pricing & RFQ

Objective: Support institutional and business buyers.

Actors: B2B Customer, Sales, Admin

Core features:

- Business profile
- GST details
- Bulk pricing
- RFQ
- PO reference
- Customer-specific pricing

Functional requirements:

FR-B2B-101 - Business accounts store organization and authorized-contact data.

FR-B2B-102 - Bulk price rules support SKU/category/customer segment.

FR-B2B-103 - RFQs have status, line items, quantities and response history.

FR-B2B-104 - B2B pricing is evaluated server-side.

Business rules: Customer-specific prices never leak to unauthorized users.

Dependencies: Users, catalog, pricing, orders, tax

Edge cases: Quote expiry; price change; unauthorized contact.

Acceptance criteria: B2B users can request quotes and convert accepted quotes to orders.

MOD-22 - Admin Dashboard

Objective: Give authorized staff operational control.

Actors: Admin, Super Admin, Staff

Core features:

- KPIs
- Product CRUD
- Orders
- Users
- Inventory
- Coupons
- Reviews

- Support
- CMS
- Reports

Functional requirements:

FR-ADM-101 - Admin navigation is permission-aware.

FR-ADM-102 - Sensitive actions require confirmation and are audited.

FR-ADM-103 - Bulk actions validate before destructive changes.

Business rules: Staff receive least-privilege access.

Dependencies: All operational modules, RBAC, audit

Edge cases: Bulk update failure; partial import; permission change mid-session.

Acceptance criteria: Staff see only authorized modules and sensitive actions are traceable.

MOD-23 - RBAC, Security & Audit

Objective: Protect business data and control operational access.

Actors: Super Admin, Admin, Staff

Core features:

- Roles
- Permissions
- Audit logs
- Session controls
- Rate limits
- Security events

Functional requirements:

FR-SEC-101 - Authorization is enforced server-side.

FR-SEC-102 - Privileged actions log actor, time, target and outcome.

FR-SEC-103 - Rate limiting protects authentication and sensitive endpoints.

FR-SEC-104 - Secrets are never committed to source control.

Business rules: Security logs exclude raw credentials, payment secrets and unnecessary personal data.

Dependencies: Identity, API, database, monitoring

Edge cases: Brute force; token theft; privilege escalation.

Acceptance criteria: Unauthorized requests fail and privileged actions are traceable.

MOD-24 - Recommendations

Objective: Provide relevant cross-sell and discovery recommendations incrementally.

Actors: Customer, Marketing, Product

Core features:

- Related products
- Frequently bought together
- Recently viewed
- Rule-based recommendations

Functional requirements:

FR-REC-101 - Initial recommendations may be rule-based using category, compatibility and purchase relationships.

FR-REC-102 - Recommendations respect product availability unless explicitly configured.

FR-REC-103 - Recommendation events are measurable.

Business rules: Start with explainable rules; introduce ML only after quality data and evaluation criteria exist.

Dependencies: Catalog, orders, inventory, analytics

Edge cases: Cold-start products; unavailable recommendations.

Acceptance criteria: Relevant available recommendations render consistently and can be measured.

8. End-to-End Workflows

8.1 Customer purchase

- Discover -> product -> select SKU -> add to cart.
- Revalidate price, stock and quantity at cart/checkout.
- Validate address, shipping, tax and coupon.
- Create idempotent payment attempt and reserve stock.
- Verify payment server-side/provider webhook.
- Confirm order, commit inventory, generate invoice/shipment and notify.
- Process shipment events through delivery.

8.2 Return/refund

- Submit return -> eligibility check -> approval/rejection.
- Create reverse logistics where required.
- Inspect and record inventory disposition.
- Initiate refund or replacement linked to original order.
- Notify customer and audit material actions.

8.3 Catalog publication

- Draft -> validate SKU/category/attributes/media -> preview -> publish -> index -> storefront.

9. Business Rules

ID	Rule
BR-001	SKU is the purchasable/inventory unit and is unique.
BR-002	Customer-facing totals are calculated or validated server-side.
BR-003	Inventory reservation is concurrency-safe.
BR-004	Orders preserve historical price, tax and item snapshots.
BR-005	Payment success requires server/provider verification.
BR-006	Coupon usage is enforced server-side and safely under concurrency.
BR-007	Privileged actions are permission-controlled and audited.
BR-008	Commercially relevant history is archived/soft-deleted rather than destructively deleted.
BR-009	Specifications are category-aware and extensible.
BR-010	B2B/customer-specific pricing is protected from unauthorized users.
BR-011	Tax/invoice rules are legally reviewed before production.
BR-012	External provider events are idempotently processed.

10. Data Model and Database Requirements

Entity	Key fields / relationships
User	id, identity, status, roles, timestamps
Organization	id, legal name, GSTIN, contacts, billing details
Address	id, owner, type, postal/tax location
Category	id, parent_id, slug, metadata, visibility
Brand	id, name, slug, metadata
Product	id, category, brand, title, description, publication state
SKU	id, product_id, code, variant data, pricing reference
AttributeDefinition	id, category scope, name, type, validation
ProductAttributeValue	product/SKU + attribute + typed value
InventoryItem	SKU + warehouse + on_hand + reserved
Cart / CartItem	user/session + SKU + quantity
Order / OrderItem	header + immutable commercial item snapshot
Payment	order + provider reference + state + amount
Shipment	order + provider + tracking + state
Return / Refund	order/payment + status + reason
Review / Question	user + product/order evidence + moderation
Coupon / Promotion	rules, validity, usage, scope
Invoice	order + tax/customer/seller snapshot
Ticket	customer + optional order/product + status/assignment
AuditLog	actor + action + entity + metadata/before-after

Use database transactions for order/payment/inventory consistency; enforce unique constraints at the database layer; use migrations and indexes based on measured query patterns; maintain timestamps and appropriate archive fields.

11. API Requirements

Domain	Representative endpoints
Auth	POST /auth/register, /login, /refresh
Catalog	GET /products, GET /products/:id, POST /admin/products
Search	GET /search?q=...&filters;=...
Cart	GET /cart, POST /cart/items, PATCH /cart/items/:id
Checkout	POST /checkout/validate, POST /checkout/order
Payments	POST /payments/create, POST /payments/webhook
Orders	GET /orders, GET /orders/:id, POST /orders/:id/cancel
Shipping	POST /shipments, GET /shipments/:id
Admin	Permission-protected /admin/* APIs

Use OpenAPI, consistent response/error formats, strict validation, pagination, idempotency keys for non-idempotent operations, correlation IDs, rate limiting and safe error messages. Webhook signatures must be verified where supported.

12. RBAC and Permissions

Capability	Customer	Support	Catalog	Inventory	Order	Finance	Marketing	Admin
Browse/purchase	Y	Y	Y	Y	Y	Y	Y	Y
Manage products	-	-	Y	-	-	-	limited	Y
Manage inventory	-	-	-	Y	-	-	-	Y
Manage orders	own	support	-	-	Y	view	-	Y
Refunds	request	request	-	-	request	Y	-	Y
CMS	-	-	limited	-	-	-	Y	Y
Coupons	-	-	-	-	-	-	Y	Y
Users/Roles	-	-	-	-	-	-	-	Y

13. Third-Party Integrations

Integration	Purpose	Design requirement
Payment gateway	Payments/refunds	Adapter + verified webhooks + reconciliation
Shipping providers	Rates/shipments/tracking	Provider abstraction + normalized states
Email	Transactional/marketing	Async jobs + templates + delivery tracking
SMS/WhatsApp	OTP/notifications	Consent-aware provider abstraction
Object storage	Images/documents	Managed buckets; signed access for protected files
Search	Catalog search	Rebuildable index; database remains source of truth
Analytics	Product/business events	Stable schema + privacy controls
Monitoring	Errors/traces	Sentry/OpenTelemetry

14. Non-Functional Requirements

ID	Requirement
NFR-PERF-001	Optimize primary customer pages for strong Core Web Vitals under representative production conditions.
NFR-PERF-002	Define latency budgets for core APIs; prioritize checkout/payment paths.
NFR-SCALE-001	Application instances support horizontal scaling where practical.
NFR-AVAIL-001	Critical commerce components avoid single-instance production dependencies.
NFR-SEC-001	Production traffic uses HTTPS/TLS.
NFR-SEC-002	Secrets are stored in managed secret/configuration systems.
NFR-SEC-003	Sensitive endpoints use rate limits and abuse controls.
NFR-DATA-001	Database backups are automated and restore-tested.
NFR-OBS-001	Logs, metrics and traces diagnose order/payment failures.
NFR-A11Y-001	Storefront targets WCAG 2.2 AA principles.
NFR-SEO-001	Public product/category pages support crawlable URLs, metadata, canonical URLs and structured data where accurate.

15. Security, Privacy and Compliance

- Apply least privilege to staff roles and enforce authorization server-side.
- Hash passwords with a modern password hashing algorithm; never store raw credentials.
- Use secure authentication/session practices and protect sensitive state-changing requests.
- Validate external input; use parameterized database access; defend against XSS, CSRF where applicable, injection and broken access control.
- Prefer tokenized/hosted payment flows; do not store raw card data unless the organization has the required compliance controls.
- Minimize personal data and define retention/deletion policies with legal guidance.
- Audit refunds, price changes, inventory adjustments, role changes and other privileged operations.
- Review applicable Indian GST, consumer protection, privacy and e-commerce obligations with qualified professionals before launch.

16. SEO and Accessibility

SEO: server-rendered/indexable product and category content, stable slugs, unique metadata, canonical URLs, XML sitemap, robots controls, accurate structured data, internal linking, pagination strategy and image alt text.

Accessibility: semantic HTML, keyboard navigation, visible focus, accessible forms/errors, meaningful labels, adequate contrast, reduced-motion consideration and responsive layouts.

17. Observability, Backups and Disaster Recovery

Area	Requirement
Logs	Structured logs with severity/correlation ID and no secrets.
Metrics	Traffic, latency, errors, queue depth, payment failures, orders and inventory exceptions.
Tracing	Trace checkout, payment, order and external integration calls.
Alerts	Payment outage, elevated 5xx, queue backlog, database health and inventory exceptions.
Backups	Automated encrypted database backups with retention policy.
Restore tests	Periodically verify backups produce a usable database.
DR	Document and test RPO/RTO targets and recovery runbooks before high-volume launch.

18. Testing and QA

Level	Scope
Unit	Pricing, tax, inventory, state transitions, validation, permissions
Integration	Database, queues, payment/shipping adapters, webhooks
API	Auth, authorization, schemas, idempotency, errors
E2E	Search -> product -> cart -> checkout -> payment -> order; returns; admin
Performance	Catalog, search, checkout and admin under representative load
Security	Dependency scanning, SAST, access-control and rate-limit tests; periodic penetration testing
Regression	Automated critical-path suite on every release

Release gate: no open P0 defects; critical checkout/payment/order scenarios pass; migration/rollback plans are reviewed; monitoring and alerts are verified.

19. Deployment and CI/CD

Recommended pipeline: pull request -> lint/typecheck/unit tests -> integration tests -> build -> security/dependency checks -> staging -> E2E -> approval -> production.

Use Docker for consistent backend/worker runtime. Version database migrations. Separate local, development, staging and production credentials, databases, storage and provider accounts.

20. Acceptance Criteria

- Customers can register/login, browse/search products, view specifications, add valid SKUs and complete checkout.
- Server-side price, tax, coupon and inventory validation prevents inconsistent totals and overselling.
- Payment events are verified and idempotently reconciled.
- Orders use controlled states and customers can view order history/tracking.
- Authorized staff can manage catalog, inventory, orders, customers and support with least privilege.
- Sensitive administrative actions are auditable.
- GST/tax/invoice behavior is reviewed and approved for the actual business model.
- Returns/refunds work for configured policies and remain linked to original records.
- Critical paths have automated tests and production monitoring.
- SEO, accessibility and performance checks meet agreed launch thresholds.

21. Roadmap and Future Scope

Phase	Scope
MVP	Storefront, auth, catalog/SKU, categories/brands, search, cart, checkout, payment, orders, shipping, admin, inventory, basic invoices and notifications
Phase 2	Returns/refunds, reviews, Q&A, CMS, marketing, analytics dashboards, advanced coupons, support tickets, B2B pricing/RFQ
Phase 3	Multi-warehouse, OpenSearch, advanced merchandising, customer-specific pricing, automation and richer analytics
Future	Recommendation/ML systems, marketplace capabilities, internationalization/tax, native mobile apps and advanced warehouse integrations

Do not introduce microservices merely for fashion. Extract a service when independent scaling, deployment, reliability isolation, team ownership or technology constraints create a measurable benefit.

22. Glossary

Term	Meaning
SKU	Stock Keeping Unit; the specific purchasable/inventory unit.
Variant	Selectable product option such as size, color, capacity or model.
RFQ	Request for Quotation; B2B process for negotiated/bulk pricing.
RBAC	Role-Based Access Control.
RPO	Recovery Point Objective - acceptable data-loss window.
RTO	Recovery Time Objective - target service restoration time.
Idempotency	Repeating an operation does not create duplicate side effects.
Webhook	Provider-to-platform HTTP event notification.
Soft delete	Archive/inactivate a record instead of physically deleting it.

End of document.